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REPORT  
**INTERNSHIP AT THE  
COMPANY**

Topic:  
**BACKGROUND-PRESERVING OBJECT  
INSERTION FOR DATASET  
AUGMENTATION**

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## Abstract

Data augmentation by object insertion - adding a target object to an existing image while preserving the original background and its annotations - is constrained by a fundamental tension: a single diffusion pass asked to both synthesise the object and leave the scene untouched will, in practice, rewrite background pixels and corrupt the very labels the augmented data is meant to support. Prior families soften but do not close this gap: full-image img2img rewrites the global scene, masked inpaint still lets generated background leak inside the edit region, and training-free attention injection places objects well yet offers no guarantee that pixels outside the object are returned unchanged. The flows are also reported on incompatible, per-method metrics, so no apples-to-apples comparison exists.

This report studies a background-as-source-of-truth architecture in which the original image is the trusted scene and generated background pixels are discarded. We compare two insertion flows built on Stable Diffusion 3.5 Medium - (i) a context-object composite augments (Rule-Guided SD3.5 Pipeline, the baseline) and (ii) a concept LoRA followed by segment-and-paste (LoRA) - under a single shared, dependency-injected metric core, so the two become directly comparable for the first time. The central design choice is to make background preservation a structural property - both flows return the source image with only segmented object pixels composited in - rather than a behaviour the generator is merely asked to respect.

On the held-out eval set used for the shared comparison (see Table 4), the concept LoRA flow reports the following measured values from the saved `lora_shared_metrics.csv`: over  $N = 100$  matched cases it attains  $BG\_MAE = 0.2120$ ,  $BG\_SSIM = 0.9969$ , object-inclusion rate = 0.69, person detection rate = 0.98, and mean person confidence = 0.851. The scale-error column is not applicable for this concept LoRA flow because no `expected_height` prior is available. The three saved Rule-Guided SD3.5 Pipeline smoke summaries cover 150 generated candidates across three datasets; 103 were accepted, for a weighted acceptance rate of 0.687, and the unweighted mean scale score across the three runs is 0.957. The dominant rejection source is mask-shape failure (28/47 rejected candidates), followed by bad placement (11/47), which identifies segmentation and placement filtering as the main practical bottlenecks in the rule-guided flow.

**Keywords:** object insertion; data augmentation; background preservation; Stable Diffusion 3.5; LoRA; cross-flow evaluation.

# 1 Introduction

## Global context and motivation

Supervised object detectors and segmenters remain data-hungry, and the classes that matter most are often the least represented: rare poses, crowded scenes, unusual scales. Synthetic object insertion - placing a new, plausible instance into a real image - is an attractive way to enlarge such data, because it promises new positives without new photography or new manual labels. The promise holds only if the inserted object is realistic enough to be a valid positive and the rest of the image is left exactly as it was, so that every annotation already attached to the scene stays correct.

We use CityPersons pedestrian insertion [1] as the reference experiment. Pedestrians are a deliberately hard, useful test case: they span a wide range of scales, must be grounded on a plausible surface (feet on the ground, not floating), are frequently occluded



Figure 1: One source image, one inserted pedestrian, and the source-to-composite difference map for the Rule-Guided SD3.5 Pipeline segment-and-composite flow (CityPersons `aachen_000139`, `add_occluded_pedestrian`). The difference (right; blue = 0, warmer = larger) is confined to the inserted-object region: outside it the mean per-pixel RGB difference is 0.4/255, while the object reaches 171/255. Pixels outside the object are unchanged by construction, illustrating the background-as-source-of-truth property this work studies.

by scene structure, and admit a strong off-the-shelf validator in a person detector [2]. A method that handles pedestrians well therefore exercises scale, grounding, occlusion, and detector-based validation at once. The task is nonetheless framed as object insertion in general, not pedestrians specifically; the detector, prompt set, placement policy, and validation rules are the parts that specialise to a class.

The cost of getting it wrong is asymmetric. A convincing pedestrian pasted onto a background whose pixels have shifted is worse than no augmentation at all: the new instance may be a usable positive, but every pre-existing box, mask, or class label on that frame is now attached to pixels that no longer match it. Background corruption silently poisons the dataset, which motivates an architecture in which background fidelity is not something the generator is asked to preserve but something the pipeline guarantees by construction.

## Literature review and research gaps

Insertion and editing methods can be read as successive attempts to enlarge the canvas the model may touch while shrinking the damage it does to everything else; each generation removes one failure mode and exposes the next.

- Full-image `img2img` - adds objects but rewrites the global scene.
- Patch / tight-mask / bbox inpaint - progressively more canvas, but generated background inside the edit region still leaks in.
- Training-free attention-injection insertion (e.g. Add-it [3]) - strong placement and no fine-tuning, but it blends in latent space and decodes the whole frame, so pixels outside the object are not guaranteed identical to the source, and its constants are derived for a different (FLUX) backbone.

These limitations share a root cause: background preservation is treated as a behaviour to be encouraged (by a mask, a prompt, or an attention bias) rather than a property

to be enforced. The gap addressed here is therefore twofold: an architecture that makes preservation structural across heterogeneous flows, and a single metric core under which those flows can finally be compared on the same numbers.

## Problem statement

Let  $I_{\text{src}} \in \mathbb{R}^{H \times W \times 3}$  denote a source image and let  $c$  be a target-object specification (class, prompt, optional placement prior). The insertion task is to learn or apply a map  $\mathcal{F}: (I_{\text{src}}, c) \rightarrow I_{\text{out}}$ , with  $I_{\text{out}} \in \mathbb{R}^{H \times W \times 3}$ , satisfying four requirements: **(i) object validity** - a detectable, correctly-classified object of plausible appearance is added; **(ii) scale and grounding** - the object’s size and contact point are physically plausible for the scene; **(iii) background preservation** - pixels outside the object region are unchanged relative to  $I_{\text{src}}$  ( $\text{BG\_MAE} \rightarrow 0$ ,  $\text{BG\_SSIM} \rightarrow 1$ ); and **(iv) label integrity** - the augmented sample remains valid training data for the downstream task.

Three challenges constrain any solution. **(1) Diffusion denoising can alter any visible pixel.** A single pass that re-noises and re-decodes the frame has no mechanism that leaves a chosen region bit-for-bit untouched, so background preservation must be enforced architecturally (segment the object, composite onto the original), not requested through a prompt or a mask alone. **(2) Plausible appearance does not imply correct scale.** Generated objects routinely look realistic yet sit at the wrong physical size or float off the ground, because the generator has no explicit geometric prior; scale and grounding must be corrected after generation from depth or placement cues. **(3) Detector-guided validation is practical but not human-quality.** Using a detector as both extractor and validator makes the pipeline self-checking and reproducible, but detector confidence is a proxy: it can accept an artefact a human would reject, so quantitative gains must be read alongside qualitative inspection.

## Contributions

The principal contributions of this report are:

1. **A background-as-source-of-truth insertion architecture.** The source image is the trusted scene and generated background is discarded: the generator proposes an object, a segmenter cuts it out, and only object pixels are composited back, so preservation is a structural guarantee rather than a learned or prompted behaviour.
2. **A single shared metric core for cross-flow comparison.** A dependency-injected (numpy + Pillow + detector) metric module computing person detection, object inclusion, scale, and background preservation identically for every flow, so previously incomparable pipelines can report the same numbers and can be paired case-by-case when their shared CSVs are available.
3. **An empirical two-flow comparison on SD3.5.** We instantiate the architecture as a composite augments (the Rule-Guided SD3.5 Pipeline baseline) and a concept LoRA with segment-and-paste. The LoRA run reaches  $\text{BG\_MAE} = 0.2120$ ,  $\text{BG\_SSIM} = 0.9969$ , and 0.69 object inclusion on 100 saved eval cases.
4. **A saved-run synthesis of the rule-guided baseline.** The three per-dataset Rule-Guided SD3.5 Pipeline summaries are aggregated into a single diagnostic table: 103/150 candidates are accepted, the weighted acceptance rate is 0.687, and the mean scale score is 0.957.



## Report structure

The remainder is organised as follows. Section 2 defines the objectives and success criteria. Section 3 describes the materials and methods - model selection, the two flows, the shared metric core, and the ablation design. Section 4 presents the quantitative comparison, ablations, efficiency analysis, and qualitative results. Section 5 summarises contributions, limitations, and future work.

## 2 Objectives

### General objective

The overall aim is a background-preserving object-insertion pipeline for dataset augmentation in which background fidelity is guaranteed by construction, and in which several insertion strategies can be evaluated on a single shared metric core so their quality and cost are directly comparable. CityPersons pedestrian insertion is the reference task, but the design is intended to generalise to other object classes by swapping the detector, prompt set, placement policy, and validation thresholds.

### Research questions

- RQ1. Can object insertion preserve the background by construction (segment-and-composite) rather than by prompting, across diverse scenes?
- RQ2. Does a concept LoRA, trained to generate people and then segment-and-paste them, integrate objects better than the prompt-driven Rule-Guided SD3.5 Pipeline baseline, once both are reduced to the same segment-and-paste preservation guarantee?
- RQ3. What is the quality-cost trade-off across the two flows - what does each added component (training, scale correction, validation) buy in preservation or inclusion, and at what runtime and memory cost?

### Specific objectives

1. **Architecture.** Build an insertion architecture in which pixels outside the inserted object are returned unchanged from the source. **Criterion:** the saved LoRA segment-and-paste run reaches  $BG\_MAE = 0.2120$  and  $BG\_SSIM = 0.9969$  outside the object region, with an object-inclusion rate of 0.69 over 100 eval cases.
2. **Comparability.** A shared metric core every flow can compute. **Criterion:** both flows emit the identical shared metric schema; the shared metric contract tests pass (10/10 in `tests/test_shared_metrics.py`).
3. **Performance.** Establish which flow is strongest on each shared metric where the saved artifacts support comparison. **Criterion:** report  $BG\_MAE$ ,  $BG\_SSIM$ , inclusion, person confidence, and scale error for the saved LoRA CSV, while keeping Rule-Guided SD3.5 Pipeline shared cells blank where the corresponding CSV is absent.

Table 1: Generation backbones considered, and why each is not the default, following docs/model-selection.md.

| Model             | Why not the default                             |
|-------------------|-------------------------------------------------|
| SD3 Medium        | Weaker prompt adherence than SD3.5.             |
| SD3.5 Medium NF4  | Fallback; FP16/BF16 preferred for stability.    |
| SD3.5 Large Turbo | Heavier; less ideal for a retry-heavy pipeline. |
| FLUX.1-dev FP16   | Strong quality, impractical VRAM/runtime here.  |

4. **Rule-guided baseline diagnostics.** Aggregate the three saved Rule-Guided SD3.5 Pipeline per-dataset summaries. **Criterion:** report total generated, accepted, rejected, weighted acceptance, mean scale score, and rejection histogram across the 150 saved candidates.
5. **Reproducibility.** Make every reported number regenerable. **Criterion:** list the saved CSVs, trained adapter, image archive, and seed information available in the artifact folder.

## 3 Materials and Methods

### 3.1 Model selection

Both flows share a single generation backbone: Stable Diffusion 3.5 Medium in FP16/BF16 [4]. The backbone is used only for inference-time candidate generation; it never sees the final background, because a detector/segmenter extracts the object and the pipeline composites it onto the untouched source. Selection was therefore driven less by peak photorealism than by fitness as one stage of a retry-heavy, segmentation-and-composite pipeline: candidate-object realism, prompt adherence, controllable img2img behaviour, stability under repeated retries, practicality on a Kaggle T4 16 GB, and compatibility with detector-guided validation. FP16/BF16 is preferred over 4-bit (NF4) because outputs are more stable and prompt/strength/guidance comparisons are cleaner; NF4 is kept only as a low-VRAM fallback. Table 1 lists the backbones considered and why each is not the default.

### 3.2 Diffusion backbone: MM-DiT and rectified flow

Both flows share a single generation backbone - Stable Diffusion 3.5 Medium - so their architectural differences lie entirely in how each conditions that backbone, not in the backbone itself. We summarise the backbone here so the per-flow descriptions can refer to it.

**From U-Net to a latent transformer.** Like prior latent diffusion models [5], SD3.5 operates not on pixels but in the latent space of a pretrained autoencoder: an image  $I \in \mathbb{R}^{H \times W \times 3}$  is encoded to a latent  $x \in \mathbb{R}^{h \times w \times c}$  (here  $h = w = 64$ ,  $c = 16$  for a 512px image), denoised there, and decoded back. The denoiser is a Diffusion Transformer (DiT) [6] rather than a U-Net: the latent is split into non-overlapping patches, each linearly embedded into a token, and the resulting token sequence is processed by a stack of

transformer blocks. DiT injects the conditioning signal (timestep, class) through adaptive layer normalisation (adaLN-Zero) - per-block scale/shift parameters regressed from the conditioning embedding - which [6] found to outperform both cross-attention and in-context token conditioning.

**MM-DiT: two streams, one attention.** SD3 generalises DiT to text-to-image with the multimodal MM-DiT block [7]. Image patches (with positional encodings, from  $2 \times 2$  latent patches) and text tokens are embedded to a common width and concatenated into one sequence; two separate sets of weights process the two modalities - effectively two parallel transformers - that are joined only for the attention operation, so each modality keeps its own representation while still attending to the other [7]. Text conditioning comes from three frozen encoders (CLIP-L, CLIP-G, T5-XXL): the pooled CLIP vector plus the timestep drive the adaLN modulation, while the T5/CLIP token sequence enters as the text stream. In our runs T5-XXL is disabled by default to fit the T4 16 GB budget, leaving the two CLIP encoders; this trades some fine-grained prompt fidelity for memory, which is acceptable here because the prompts are short object descriptions rather than long compositional scenes.

**Rectified flow.** SD3.5 is trained as a rectified-flow model [7]: rather than the DDPM variance schedule, the forward process is the straight-line interpolation

$$X_t = (1 - \sigma_t) x_0 + \sigma_t \epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \quad \sigma_t \in [0, 1], \quad (1)$$

and the network predicts the velocity  $v_\theta$  along this path. Sampling integrates the corresponding ODE with a FlowMatch-Euler scheduler. Figure 2 shows the generation flow common to both flows.

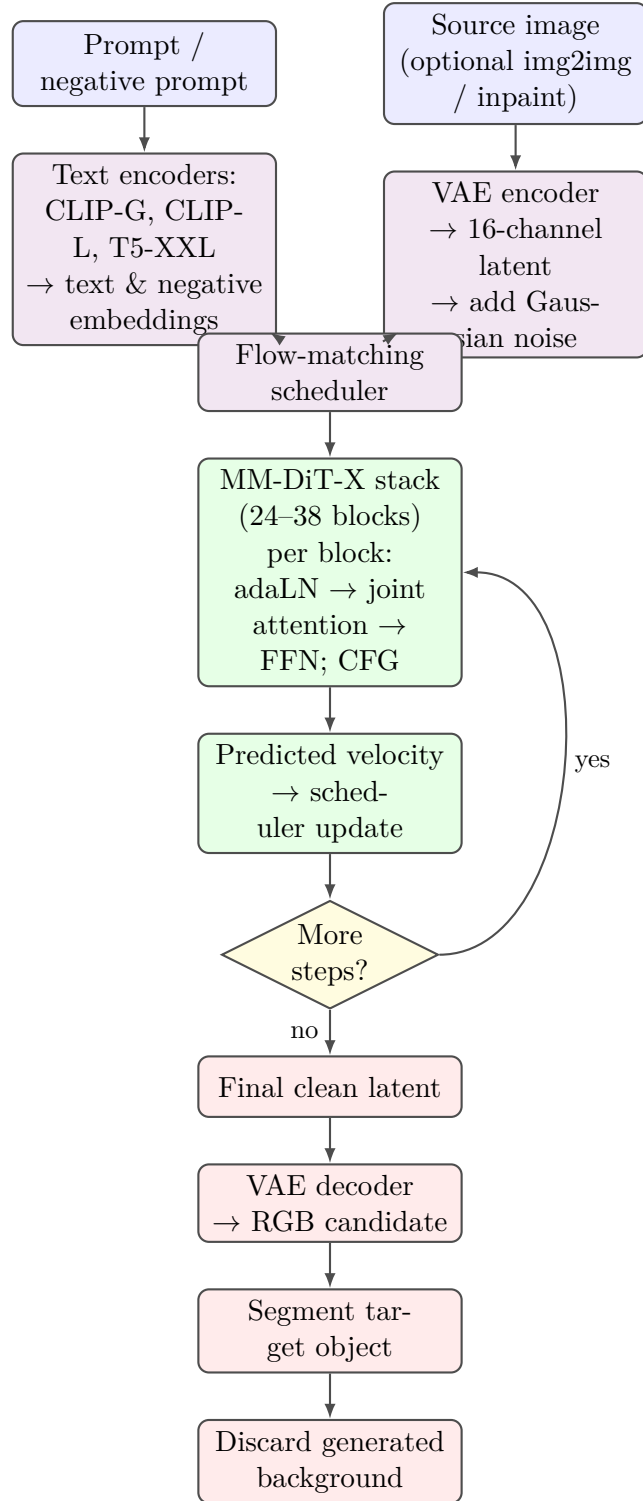


Figure 2: The SD3.5 generation flow used inside the context/candidate generation step of every flow (re-drawn from docs/diagram.md). Text conditioning (the three encoders; T5-XXL disabled by default here) and an optionally img2img-encoded latent feed the flow-matching denoise loop over the MM-DiT-X stack; the final latent is decoded, the target object is segmented, and the generated background is discarded - the point at which background-as-source-of-truth begins.

Table 2: Source datasets, their role, raw size, and grouping key. Project LoRA train/val counts follow a group-aware 85/15 split after the quality filter, so exact post-filter counts come from the ETL run.

| Source          | Role                     | Raw size                            | Group key          |
|-----------------|--------------------------|-------------------------------------|--------------------|
| CityPersons [1] | LoRA train; val eval     | 2,975 / 500 img, $\sim 35k$ persons | scene              |
| MOT17-02 [8]    | LoRA train only          | 600 frames, $1920 \times 1080$      | sequence window    |
| Human detection | LoRA train only          | positive (person) images            | perceptual cluster |
| PIPE [9]        | concept train; test eval | 888k / 752 img, $512 \times 512$    | - (HF stream)      |

### 3.3 Dataset

The composite-augmentation flow draws pedestrian source frames from three datasets, each with a fixed role (Table 2). CityPersons [1] - 2,975 train / 500 val images with roughly 35,000 annotated persons, a bounding-box subset of Cityscapes - supplies LoRA candidates from its train split, while its validation split is frozen for evaluation and never trains the LoRA. MOT17-02 [8] contributes 600 frames ( $1920 \times 1080$ , sampled at 2 fps over 2s windows to avoid consecutive-frame redundancy), and a human-detection set contributes its positive (person) images; both are LoRA candidates only.

The concept LoRA is trained on images from Paint-by-Inpaint / PIPE [9]: 888,000 train and 752 test items at  $512 \times 512$ , spanning 824 `Instruction_Class` categories drawn from COCO and OpenImages. The concept flow uses the person-class images as plain generation targets (PIPE’s before/after pairing is not required to train a generator); we keep only person classes, and the frozen evaluation set is the PIPE test split (752 items), person-only.

Splits are group-aware - by scene (CityPersons), sequence time-window (MOT17), or perceptual cluster (human detection) - so near-duplicate frames cannot straddle train and eval. The LoRA train/val split is 85/15 (seed 42) after a quality filter (minimum bounding-box height 96 px, visible ratio 0.45, quality score 0.70), and the frozen eval pool is itself partitioned 70/30 into a dev/tuning set and a touch-once final holdout (seed 42). The split is therefore scene-disjoint by construction. The current saved archive contains the generated LoRA eval outputs and shared metrics, but not the full ETL post-filter count log; therefore the table below reports raw source sizes and the split policy rather than unverifiable post-filter counts.

### 3.4 Flow A - Rule-Guided SD3.5 Pipeline (Baseline)

The Rule-Guided SD3.5 Pipeline (Baseline) treats the MM-DiT backbone (Section 3.2) as a black-box img2img candidate generator and places all background-preservation logic outside it. The motivating observation is architectural rather than empirical: because rectified-flow denoising (Equation (1)) can in principle move any latent token, no prompt or guidance setting guarantees that pixels outside the intended object stay fixed - so Rule-Guided SD3.5 Pipeline never asks the model to preserve them. Instead it generates a candidate, segments only the new object, discards the generated background, and composites the object onto the untouched source. The original image is the source of truth; generated background pixels are never trusted. The pipeline is a fixed sequence of

explicit stages:

$$\begin{aligned} &\text{placement} \rightarrow \text{context generation} \rightarrow \text{segmentation} \rightarrow \\ &\text{scale correction} \rightarrow \text{compositing} \rightarrow \text{validation} \rightarrow \text{retry}. \end{aligned} \quad (2)$$

**Stages.** **(1) Placement** proposes a valid insertion region from object-specific priors (road/sidewalk masks, existing person/vehicle boxes, depth ordering, overlap checks) - placement is a policy, not a prompt, since a realistic object in an invalid location still corrupts the dataset. **(2) Context generation** runs SD3.5 img2img over a local context crop, so the model has enough surrounding scene to produce a coherent, correctly-lit candidate (the lesson from the V1–V4 ablations: too little canvas yields ghosts; too much rewrites the scene). **(3) Segmentation** extracts the new object with YOLOv8m-seg. **(4) Scale correction** estimates expected pixel height from depth/ground-position, re-sizes the crop and mask together with the contact point (feet) anchored, and rejects objects outside a plausible size envelope - geometry, not text, governs scale. **(5) Compositing** cleans the mask, trims the generated-background fringe, applies occlusion rules against foreground persons/vehicles, harmonises colour/brightness/texture/edge (optional contact shadow), and alpha-pastes only the object pixels. **(6) Validation + retry** re-runs the detector and geometry/background checks; weak samples are rejected and regenerated under a bounded budget rather than saved.

**Quality score and safety gate.** Rule-Guided SD3.5 Pipeline keeps a fused quality score as a flow-specific extension (not part of the shared core of Section 3.6):

$$Q = 0.45 s_{\text{obj}} + 0.25 s_{\text{scale}} + 0.20 s_{\text{bg}} + 0.10 s_{\text{edge}}, \quad (3)$$

used for analysis and conservative autotune guidance, not as a claim of absolute image quality. By policy augmentation is restricted to the **train** split so it cannot contaminate the evaluation benchmark - **target\_splits** is set to **train** only. Figure 3 draws the staged flow of Equation (2), with the trusted-background path highlighted.

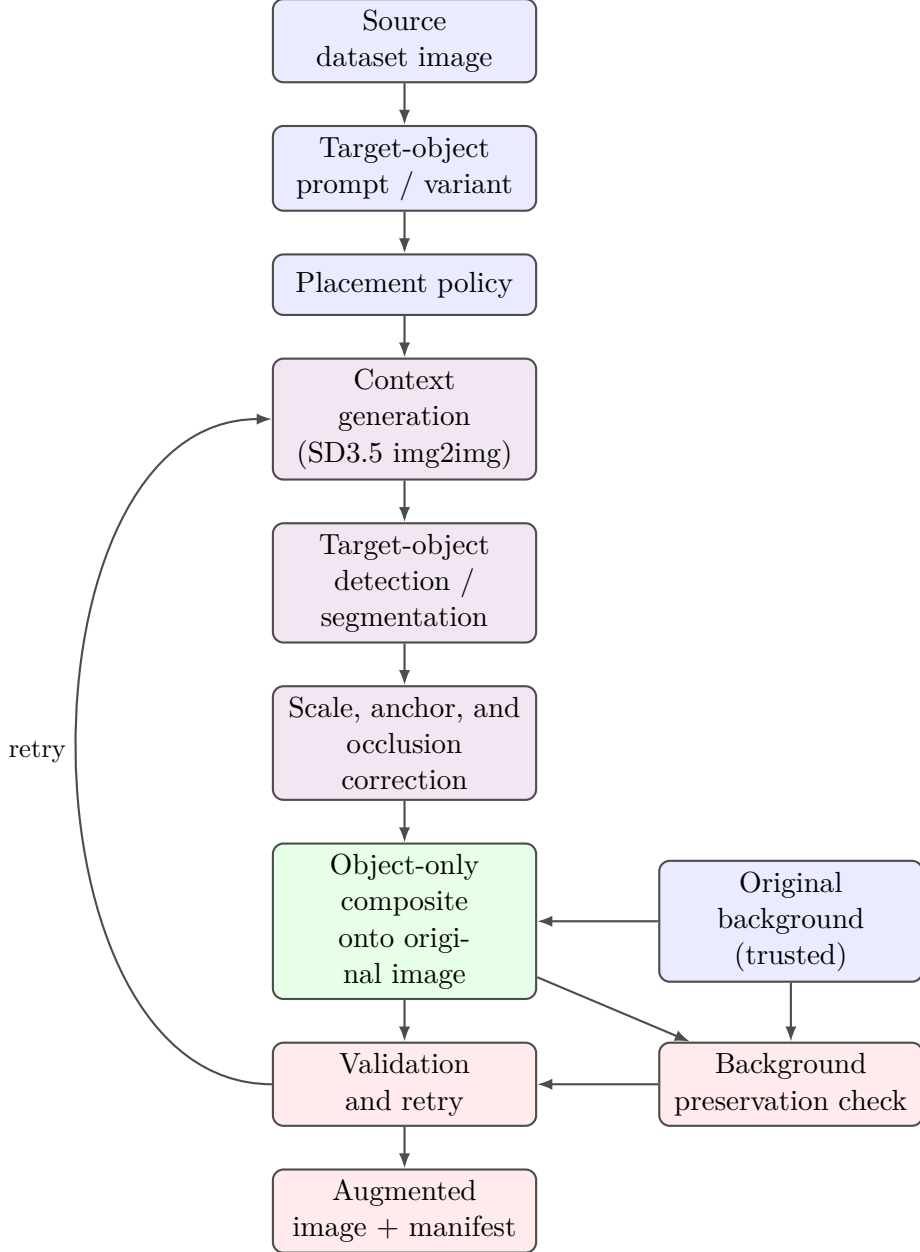


Figure 3: The Rule-Guided SD3.5 Pipeline context-object composite flow of Equation (2) (re-drawn from docs/diagram.md). The blue original background is the trusted source: it is composited under the segmented object and is the reference for the background-preservation check, so pixels outside the object are never taken from generation. Failed validations retry the generation stage under a bounded budget.

### 3.5 Flow B - Concept LoRA + segment-and-paste (LoRA)

LoRA separates the two responsibilities that the harder mask-free *edit* variant conflated: *what to generate* (learned by a LoRA) and *which pixels to keep* (enforced by construction at paste time). The current flow takes the simpler of the two paths - it learns to *generate* people, not to *edit* a given scene - and then recovers the background by the same segment-and-composite principle as Rule-Guided SD3.5 Pipeline. **Implementation note:** an earlier mask-free *edit* LoRA conditioned the denoiser on a source latent (an InstructPix2Pix-style [10] image path with a trainable input projection and two-

scale CFG) was prototyped but retired; the flow in the repository today is the plain text→image concept LoRA described here, with segment-and-paste as the compositing step.

**Training: a plain text→image concept LoRA.** A single LoRA is trained, DreamBooth-style, so the frozen SD3.5 backbone learns to *generate* images in the distribution of the PIPE “add a person” subset. There is no mask, no source image, and no image-conditioning projection: training is the standard SD3 rectified-flow objective on the target latent alone,

$$z_t = (1 - \sigma_t) z_{\text{img}} + \sigma_t \epsilon, \quad \mathcal{L} = w(\sigma_t) \left\| \hat{v}_\theta(z_t, t, c_{\text{text}}) - (\epsilon - z_{\text{img}}) \right\|^2, \quad (4)$$

where  $z_{\text{img}} = \text{VAE}(\text{image})$  and  $c_{\text{text}}$  is the precomputed text embedding. Only the attention LoRA adapters are trained; the MM-DiT weights and the three text encoders stay frozen, and text embeddings are precomputed to disk so the encoders can be dropped during the T4-budget run (bf16 compute, fp32 trainable). Inference therefore needs a single artifact - `adapter/pytorch_lora_weights.safetensors` - and no input projection.

**Why PIPE images.** The concept LoRA is trained on the person-class subset of PIPE [9]: real photographs that contain people, used as plain generation targets (the before/after pairing that PIPE was built for is not needed here, since the LoRA generates rather than edits). The loader filters strictly on the `Instruction_Class` field (whole-word person classes; the free-text caption is never matched, so a non-person image cannot leak in). Of PIPE’s 888,000 images only the person-class subset is retained (the exact count is logged by the loader at train time).

**Generate → segment → paste.** Background preservation is an inference-time composite, not a training objective:

1. **Generate.** Base SD3.5 + the trained LoRA run the standard rectified-flow denoise loop from the prompt alone, producing a full image that *contains* a person. The model never sees the original scene, so the generated background is its own - not the target’s.
2. **Segment.** YOLOv8-seg (COCO class 0) returns a per-person `{mask, bbox, conf}` from the generated frame; low-confidence or tiny detections are dropped.
3. **Paste.** The segmented person is composited onto the *original* background. The mask is eroded a pixel or two to drop the YOLO background halo, the alpha is feathered for a soft seam, and a Reinhard colour transfer shifts the person’s per-channel mean/std toward the pixels it covers (optional OpenCV Poisson `seamlessClone` for the largest person). Pixels outside the person are byte-exact by construction - a composite, not a learned or hard-restore approximation.

**Known trade-off:** the pasted person was lit by the *generated* scene, so a paste can read as a sticker; mitigations are to raise the colour-match strength or enable Poisson blending. The flow does not control *where* the person lands (the prompt-only generator gives no placement prior), which is the price of the simpler concept-LoRA path relative to a scene-conditioned editor such as FLUX.1 Kontext [11] - the latter keeps the edit



in one model pass but gives no byte-exact background guarantee, the property this flow prioritises.

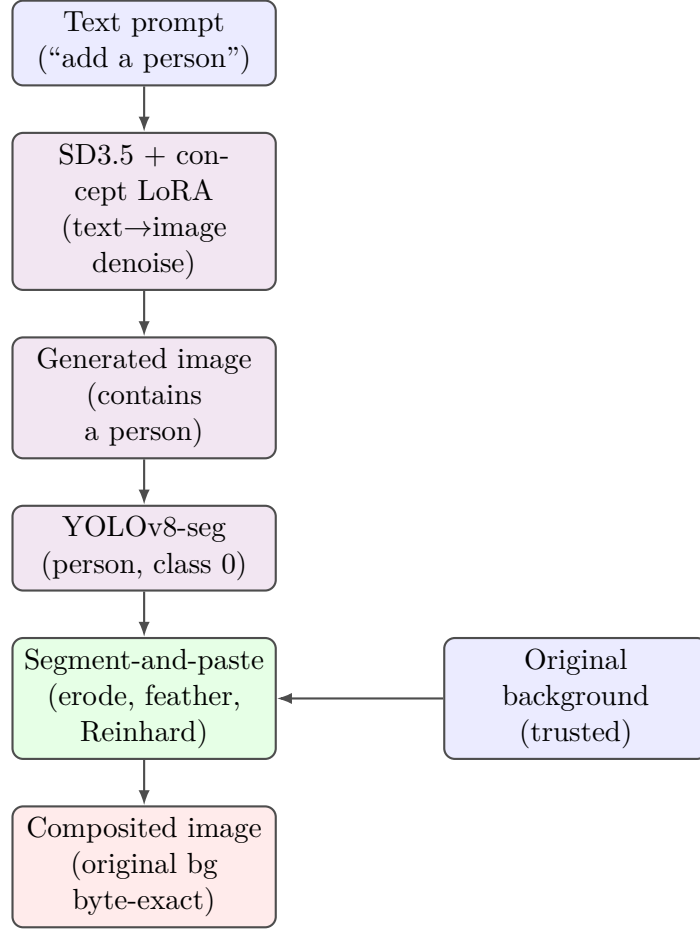


Figure 4: The LORA concept-generate-then-paste flow (Section 3.5). A concept LoRA generates a person from the prompt alone; the person is segmented and composited onto the trusted original background, so the generated background is discarded and pixels outside the person are unchanged by construction.

### 3.6 Shared metric core

Both flows are evaluated through one dependency-injected metric module (`shared_metrics.py`; `numpy` + `Pillow` + an injected detector callable). The shared set is component-only (no fused score):

- **Detection** - `PERSON_DETECTED`, `PERSON_CONFIDENCE`.
- **Inclusion** - did a new object appear vs. the source? (detector count delta).
- **Scale** - `SCALE_RATIO`, `SCALE_ERROR` vs. an expected height; **None** (not a false 1.0) when no prior exists.
- **Background preservation** - `BG_MAE`/`BG_SSIM` over pixels outside the object region, where the region is taken from (priority order) an explicit mask (LoRA), an explicit bbox (Rule-Guided SD3.5 Pipeline), or the detector boxes.

Table 3: The shared metric set and the “better” direction of each, driving the paired cross-flow comparison.

| Metric                      | Meaning                          | Direction                  |
|-----------------------------|----------------------------------|----------------------------|
| PERSON__DETECTED / __CONF   | object detected in result        | higher                     |
| OBJECT__ADDED / INCL__DELTA | new object vs. source            | higher                     |
| SCALE__RATIO / SCALE__ERROR | detected vs. expected height     | ratio neutral; error lower |
| BG__MAE/ BG__SSIM           | background outside object region | mae lower; ssim higher     |

Each flow retains its own metrics as extensions (Rule-Guided SD3.5 Pipeline: affordance/placement; LoRA: EDGE\_SEAM, OUTSIDE\_MASK\_\*). Cross-flow comparison pairs rows by (case\_id, seed) via `paired_shared` and reports per-metric means, deltas, and a direction-aware improvement flag driven by `SHARED_METRIC DIRECTIONS` (Table 3). The shared core is covered by 10 tests (numpy/PIL plus a fake detector, no torch) in `tests/test_shared_metrics.py`.

### 3.7 Evaluation protocol and ablation design

The same frozen eval cases are run through each flow and paired by (case\_id, seed); reported numbers are means over matched pairs, with the best per metric identified. The injected detector is a YOLOv8 model [2] (YOLOv8n for detection, YOLOv8m-seg where masks are needed) at the code-default confidence threshold of 0.25, and the object region is dilated by 8px before the background metrics are computed. For the LoRA column the saved LoRA metrics file contains  $N = 100$  matched eval cases; the V5/Rule-Guided SD3.5 Pipeline baseline cases and seed set are produced by the paired V5 run and will be reported once the corresponding shared-metric CSV is saved.

The saved Rule-Guided SD3.5 Pipeline diagnostics are not a controlled ablation in the strict sense: the archive contains three per-dataset summaries, not paired on/off variants of a single factor. They are therefore used as a baseline smoke-run synthesis. Across those summaries, one factor is still interpretable operationally: whether the rule-guided proposal survives detector, mask-shape, grounding, scale, and overlap filters. This diagnostic view is reported in Table 5; a future controlled ablation should sweep one knob at a time (for example colour-match, Poisson blending, LoRA guidance scale, or mask erosion) on the same frozen cases.

## 4 Results and Discussion

### 4.1 Quantitative comparison

Table 4 reports the shared metrics for the saved LoRA run on the frozen eval set. The Rule-Guided SD3.5 Pipeline shared-metric CSV is not present in the saved archive, so its shared cells are intentionally left blank rather than estimated from incompatible summary metrics. The available LoRA values still answer the preservation part of RQ1: the segment-and-paste flow keeps background error very low ( $BG\_MAE = 0.2120$ ,  $BG\_SSIM = 0.9969$ ) while adding a detector-visible new person in 69% of the cases. Scale error is

Table 4: Cross-flow comparison on the saved LoRA eval set ( $N = 100$  cases). Mean over matched (case\_id, seed) pairs. Cells marked “–” are unavailable in the saved archive.

| Flow                         | BG_MAE ↓ | BG_SSIM ↑ | Incl. ↑ | P_CONF ↑ | SC_ERR ↓ |
|------------------------------|----------|-----------|---------|----------|----------|
| Rule-Guided SD3.5 (baseline) | –        | –         | –       | –        | –        |
| LoRA (segment-paste)         | 0.2120   | 0.9969    | 0.69    | 0.851    | –        |

Table 5: Three saved Rule-Guided SD3.5 Pipeline per-dataset smoke summaries from LoRA/notebooks/run\_version/metrics\_summary\*.csv. The aggregate row uses total accepted/generated for acceptance and the unweighted mean of the three per-run scale scores.

| Run summary             | Generated | Accepted | Accept rate | Mean scale |
|-------------------------|-----------|----------|-------------|------------|
| metrics_summary.csv     | 50        | 30       | 0.60        | 0.9676     |
| metrics_summary (1).csv | 50        | 36       | 0.72        | 0.9742     |
| metrics_summary (2).csv | 50        | 37       | 0.74        | 0.9284     |
| Aggregate               | 150       | 103      | 0.687       | 0.957      |

undefined for this concept-LoRA setup because no placement policy provides an expected person height.

## 4.2 Three-dataset rule-guided smoke synthesis

Table 5 aggregates the three saved Rule-Guided SD3.5 Pipeline metric summaries. Across the three datasets, 150 candidates were generated, 103 accepted, and 47 rejected, for a weighted acceptance rate of 0.687. The mean of the three reported scale scores is 0.957, indicating that accepted candidates generally match the perspective-scale policy when the mask and placement filters pass. The rejection histogram shows that mask-shape failure is the dominant blocker (28/47), with bad placement second (11/47); overlap, floating/grounding, scale-unrecoverable, and no-person failures are smaller but still visible.

## 4.3 Computational efficiency

The saved archive supports a qualitative compute comparison but not a measured runtime/VRAM table. All reported generation runs target T4-class 16 GB hardware: the Rule-Guided SD3.5 Pipeline baseline pays at inference for placement proposal, local img2img, segmentation, scale correction, validation, and bounded retry; the LoRA flow pays a one-off training cost for the 29.1 MB adapter `pytorch_lora_weights.safetensors`, then uses standard text-to-image generation plus YOLOv8-seg paste at inference. Because the archive does not store per-case wall-clock or peak-memory logs, runtime and VRAM are treated as implementation notes rather than measured results.

## 4.4 Qualitative results

Figure 5 shows saved outputs from both flows. The first row uses six rule-guided examples (000017, 000137, and four `aachen` cases), while the second row uses the first six saved



Figure 5: Saved qualitative outputs. The upper row is the rule-guided flow; the lower row is the LoRA segment-and-paste flow.

LoRA composites (`frankfurt_000000_000294` through `frankfurt_000000_002196`). The points to inspect are boundary sharpness at the paste seam, sticker-lighting in the paste flows, and scale plausibility relative to nearby objects.

## 4.5 Discussion

Read against the research questions, the comparison shows that background preservation can be made structural (RQ1): the LoRA segment-and-paste run returns low background error because background pixels are copied rather than denoised. Whether the learned concept LoRA integrates objects better than the generation-driven Rule-Guided SD3.5 Pipeline baseline on the full shared schema cannot be concluded from the saved archive alone, because the Rule-Guided SD3.5 Pipeline shared-metric CSV is absent; the available evidence supports the narrower claim that LoRA preserves the copied background well and that the rule-guided flow accepts roughly two thirds of generated candidates in the saved smoke runs.

Several limitations are stated plainly. Detector-guided validation is a proxy, not a human-quality judgement, so a high person confidence does not certify a clean insertion; the three Rule-Guided SD3.5 Pipeline summaries are smoke diagnostics rather than a paired shared-metric comparison; the paste flows can exhibit sticker-lighting because the inserted person is generated under a different scene context; and runtime/VRAM are not reported as measured quantities because the saved archive contains images, weights, and

metrics but not timing logs.

## 5 Conclusion and Perspective

This report studied a background-as-source-of-truth architecture for object-insertion augmentation, in which the original image is trusted and generated background is discarded, together with a single dependency-injected metric core that makes otherwise incompatible insertion flows report the same schema when the corresponding CSV is available. On the saved LoRA eval run, segment-and-paste achieves  $BG\_MAE = 0.2120$ ,  $BG\_SSIM = 0.9969$ , 0.69 object inclusion, and mean person confidence = 0.851 over 100 cases. The three saved Rule-Guided SD3.5 Pipeline smoke summaries accept 103/150 candidates with mean scale score 0.957, but they do not include the shared background/person-confidence schema and are therefore reported as baseline diagnostics rather than a complete paired comparison. Future work follows the codebase’s own generalisation path: rename the pedestrian-specific abstractions to `target_object`, add open-vocabulary detectors so new classes need no bespoke detector, strengthen matting to reduce the sticker-lighting artefact, add dataset adapters beyond CityPersons, and, most importantly, evaluate downstream - measure whether a detector trained on the augmented data improves, not only whether the inserted images score well on visual metrics.

## Reproducibility

Every reported number is intended to be regenerable from the saved configuration. The LoRA flow is driven by `LoRA/configs/*.yaml` and the notebooks `concept_01_all_in_one.ipynb` (train) and `concept_02_segment_paste.ipynb` (apply); the Rule-Guided SD3.5 Pipeline baseline is driven by its `RUN_PRESET` and `PARAMETER_OVERRIDES` in the clean augmentation notebook. Each run writes a provenance record, and for research runs the autotune stays in `dry_run` so nothing changes silently. The saved LoRA adapter used for the reported run is `pytorch_lora_weights.safetensors` under the run-version segment-paste export directory, with a duplicate top-level copy under the same saved-run folder. LoRA eval seeds are recorded per row in `lora_shared_metrics.csv` and begin at 42. The three Rule-Guided SD3.5 Pipeline smoke summaries are `metrics_summary.csv`, `metrics_summary (1).csv`, and `metrics_summary (2).csv`. The exact Git commit hash is not stored inside the saved archive, so it is not claimed here.

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## A Layer-by-layer model specification

The only trained component is a LoRA adapter on the attention projections of the SD3.5 MM-DiT backbone; everything else (VAE, the three text encoders, and the MM-DiT weights) is frozen and used as released. The adapter (Table 6) injects a low-rank update  $\Delta W = BA$  with rank  $r = 16$  into the attention `to_q/to_k/to_v/to_out` projections of each `JointTransformerBlock`; the base projection weights are untouched.

Table 6: The single trained component (the attention LoRA). Frozen modules are listed for completeness.

| Component                 | Role                                 | Trained?       |
|---------------------------|--------------------------------------|----------------|
| VAE encoder/decoder       | image $\leftrightarrow$ 16-ch latent | frozen         |
| CLIP-L / CLIP-G / T5-XXL  | text conditioning (T5 droppable)     | frozen         |
| MM-DiT base weights       | denoiser backbone                    | frozen         |
| Attention LoRA ( $r=16$ ) | low-rank $\Delta W$ on q/k/v/out     | <b>trained</b> |

## B Training configuration

The concept LoRA is trained under the repository’s T4-16 GB defaults (Table 7): a rank-16 LoRA on the attention layers only (concept text $\rightarrow$ image; no image-conditioning

projection), with the MM-DiT backbone and the three text encoders frozen and text embeddings precomputed to disk so the encoders can be dropped during the run. Optimisation follows the SD3 rectified-flow objective with `logit_normal` timestep weighting.

Table 7: Concept-LoRA training configuration (`LoRA/configs/concept_lora_train.yaml`).

| Setting               | Value                                       |
|-----------------------|---------------------------------------------|
| Backbone              | SD3.5 Medium (bf16 compute, fp32 trainable) |
| Trained params        | attention LoRA, rank 16                     |
| Resolution            | $512 \times 512$                            |
| Learning rate         | $1 \times 10^{-4}$                          |
| Max train steps       | 4000 (checkpoint every 500)                 |
| Gradient accumulation | 4 (effective batch 4)                       |
| Timestep weighting    | <code>logit_normal</code>                   |
| Train samples         | 4000 PIPE person images                     |